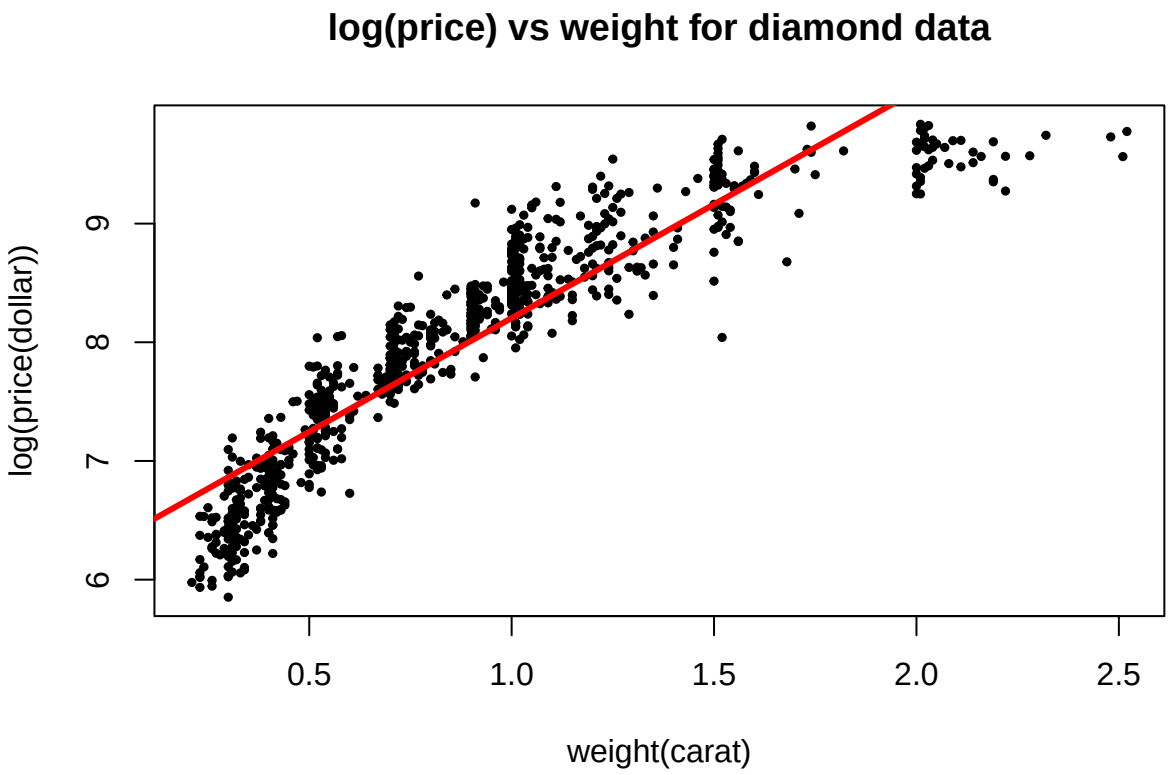
st231 a2

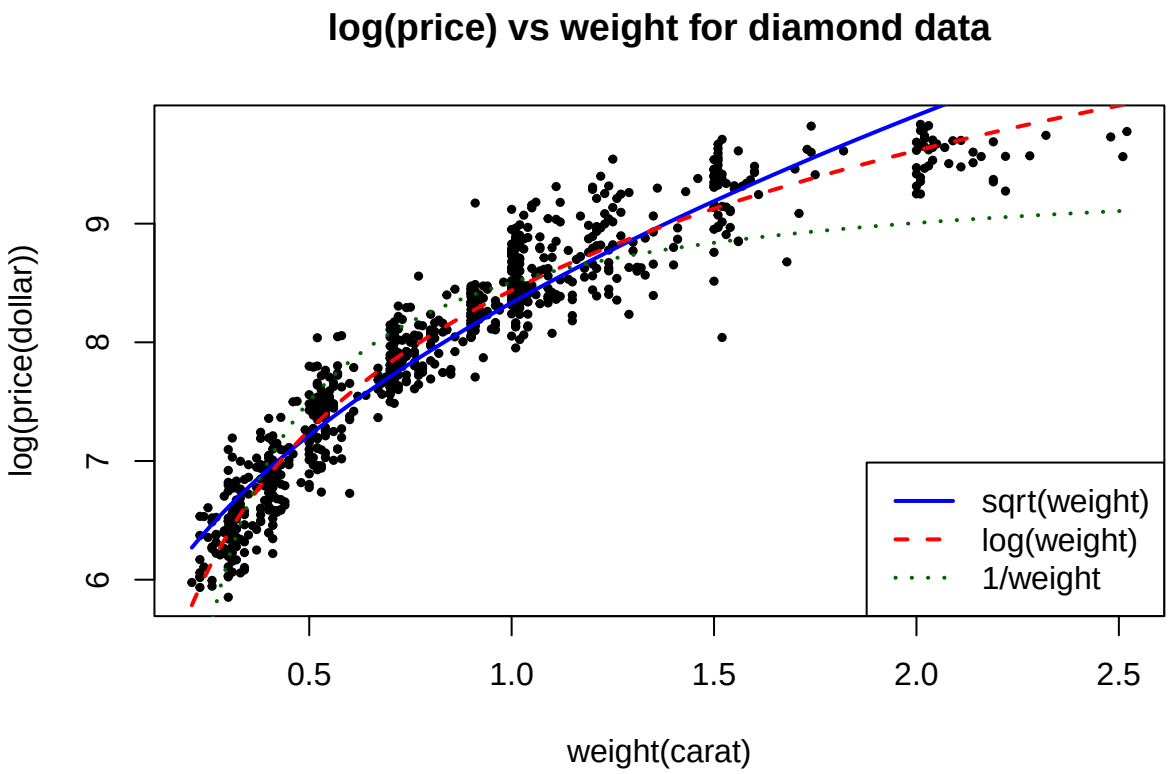
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1.The assumption of a linear relationship is not reasonable because the shape of the scatters is a positive concave curve. Linear relationship overestimates the log(price) for weight$<$ 0.5 carats and weight$>$ 2.0 carats.

2.From the first graph, the relationship is positive but concave.On the Tukey-Mosteller grid, this corresponds to a bulge pointing toward the top-left quadrant (Quadrant 2). Then we should transform weight using a power p$<$1.



3. Model 1: log(price) $\sim \sqrt{\text{weight}}$ (Solid Blue Line) It starts off following the data well but eventually begins to overshoot the points as weight increases beyond 1.5.
Model 2: log(price) $\sim$ log(weight) (Dashed Red Line) This curve tracks the central density of the data points very closely across the entire range.
Model 3: log(price) $\sim 1/\text{weight}$ (Dotted Green Line) It fits the very small weights okay,but levels off far too early. By the time weight reaches 1.5, the curve is well below almost all the data points.
Therefore, Model 2: log(price) $\sim$ log(weight) is the most appropriate.